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gradient found in Eq. (7). The mathematical details are given in [Sisini et al.(2015b)]. 3
Figure 3: Calculated velocity trace together with the experimental one obtained by the
Doppler examination. RESULTS EXAMPLE Pressure waves velocity calculation The measured delay (t_{aQP}) was 0.14 s and the distance LG was 23 cm. As a consequence the velocity c was 164 cm/s.
Compliance calculation The minimum value of the CSA (CSA_x) during the cardiac cycle was
0.2 cm² and the compliance for unit of length was $9.8 \cdot 10^{-3}$ cm²/mmHg. Flow velocity
calculation $w(t, z)$ The velocity was calculated following Eq. (1) and its trace is shown in
Fig. 3 together with the experimental trace obtained by the Doppler examination. The two
traces in agreement, nevertheless, since the velocity $w(t, G + l)$ was calculated up to an
additive constant, only the wave amplitude has physical meaning whereas its shift along the
velocity axis is not significant. ACKNOWLEDGEMENT The author wants to thank Eleuterio Toro for
his critical comments to the paper and Giacomo Gadda, Valentina Tavoni and Valentina Sisini to
have kindly reviewed the manuscript. REFERENCES [Applefeld(1990)] Applefeld MM. The Jugular
Venous Pressure and Pulse Contour. In: Source Clinical Methods: The History, Physical, and
Laboratory Examinations. Boston: Butterworths, 1990. chapter 19. [Beulen et al.(2011)] Beulen,
B. et al. Toward noninvasive blood pressure assessment in arteries by using ultrasound.
Ultrasound in Medicine; 2011: 37(5), 788-797. [Kalmanson et al. (1972)] Kalmanson D, Veyrat C,
Deraï C, Savier CH, Berkman M, Chiche P. Non-invasive technique for diagnosing